WORKED SOLUTION

STEP I 2002, Question 1

Pure • Circles and Ellipses

> **STEP I 2002, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/02-s1-q1

The key idea

Find where the two ellipses actually meet. Both equations involve y only through y^2, so eliminating y^2 leaves a quadratic in x, and it turns out that the ellipses cross at just two real points. A circle through two given points has its centre on their perpendicular bisector, and that pins the equation down to a one-parameter family.

Solution

Finding the points of intersection

Expanding, the two ellipses are

$$x^2 + 4x + 2y^2 = 14,$$
$$9x^2 - 18x + 16y^2 = 16.$$

Multiply the first by 8 and subtract the second, which removes y^2:

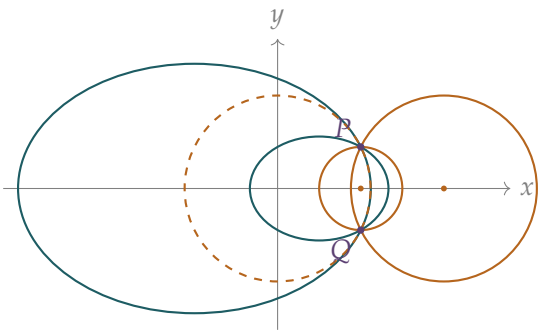
$$8x^2 + 32x - 9x^2 + 18x = 112 - 16 \quad \Longrightarrow \quad x^2 - 50x + 96 = 0 \quad \Longrightarrow \quad (x-2)(x-48) = 0.$$

> **Ask yourself** Why can't we just keep both roots?

Each root still has to give a real y. From the first ellipse, $2y^2 = 14 - x^2 - 4x$.

- $x = 48$: $2y^2 = 14 - 2304 - 192 < 0$, so there is no real point. (In any case the first ellipse only covers $-2 - 3\sqrt{2} \le x \le -2 + 3\sqrt{2}$.)
- $x = 2$: $2y^2 = 14 - 4 - 8 = 2$, so $y = \pm 1$.

So the ellipses meet in exactly two points, $P = (2, 1)$ and $Q = (2, -1)$. (Both satisfy the second equation as well: $9 + 16 = 25$.)



The figure shows the two ellipses (teal) and three of the circles through P and Q (amber), with centres $(0,0)$, $(2,0)$ and $(4,0)$.

The circles through P and Q

The centre of any circle through P and Q is equidistant from them, so it lies on the perpendicular bisector of PQ. The segment PQ is vertical with midpoint $(2,0)$, so its perpendicular bisector is the x-axis. Hence the centre is $(a,0)$ for some real a, and the radius is

$$r^2 = (2-a)^2 + 1^2.$$

The circle is therefore

$$\begin{aligned}(x-a)^2 + y^2 &= (2-a)^2 + 1\\ x^2 - 2ax + a^2 + y^2 &= 4 - 4a + a^2 + 1\\ x^2 - 2ax + y^2 &= 5 - 4a,\end{aligned}$$

which is the required form.

Conversely, for every real a this equation is a genuine circle, centre $(a,0)$ and radius $\sqrt{(a-2)^2+1} > 0$, and it passes through $(2,\pm 1)$ since $4 - 4a + 1 = 5 - 4a$. So the family is exactly the set of circles through the points of intersection.

Ask yourself What about the usual trick of taking a combination of the two equations?

Combining $7\times$(first) $-$ (second) makes the x^2 and y^2 coefficients equal and gives $x^2 - 23x + y^2 + 41 = 0$, which is the member $a = \frac{23}{2}$ of our family. But it is only one circle. Combinations of the two equations are the conics through all four (complex) common points, including the two non-real ones with $x = 48$, while a circle only has to pass through the two real points. That is why the direct approach is needed here.

Checks

- $a = 2$ gives $x^2 - 4x + y^2 = -3$, that is $(x-2)^2 + y^2 = 1$: the circle with PQ as diameter, as it should be.

- $a = 0$ gives $x^2 + y^2 = 5$, and indeed $2^2 + 1^2 = 5$.
- A numerical scan of $-6 \leq x \leq 6$ for points lying on both ellipses finds only $x = 2$. At $(2, 1)$ the normals to the ellipses are along $(8, 4)$ and $(18, 32)$, which are not parallel, so the ellipses genuinely cross there rather than touching.

Common slips

- Keeping $x = 48$ without checking that y is real.
- Producing the single circle from a combination of the ellipse equations and stopping there: the question asks about *any* circle through the points.

Beyond STEP

The combination $\lambda E_1 + \mu E_2 = 0$ of two conics is called a *pencil*: it consists of the conics through their four common points, counted in the complex projective plane. This question is a neat reminder that some of those points may be non-real, and that circles are special (they all pass through two fixed complex "circular points at infinity"). These ideas belong to projective and algebraic geometry; a friendly first look at plane curves and conics in this spirit is Reid, *Undergraduate Algebraic Geometry*, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.